به نام خدا

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پاسخ تمرین جلسه دوم

برنامه ای بنویسید که عدد ۰ تا ۹۹۹۹ را بشمارد و روی سون سگمنت چهارتایی نمایش دهد و بعد از اینکه به ۹۹۹۹ رسید مجددا این شمارش تکرار شود.

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Notes | Az2-Tamrin.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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Project : Session2-Exercise
Version :
Date    : 10/7/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>
#include <delay.h>

void SevenSegment();
void DigitSeperation(int i);

// Declare your global variables here
int sevenseg[] = {0x3f, 0x06, 0x5b, 0x4f, 0x66, 0x6d, 0x7d, 0x07, 0x7f, 0x6f};
int i,digit[4];

void main(void)
{
```

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```
// Declare your local variables here

// Input/Output Ports initialization
// Port A initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);

// Port B initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=Out Bit2=Out Bit1=Out Bit0=Out
DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (1<<DDB3) | (1<<DDB2) | (1<<DDB1) | (1<<DDB0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);

// Port C initialization
// Function: Bit7=Out Bit6=Out Bit5=Out Bit4=Out Bit3=Out Bit2=Out Bit1=Out Bit0=Out
DDRC=(1<<DDC7) | (1<<DDC6) | (1<<DDC5) | (1<<DDC4) | (1<<DDC3) | (1<<DDC2) | (1<<DDC1) | (1<<DDC0);
// State: Bit7=0 Bit6=0 Bit5=0 Bit4=0 Bit3=0 Bit2=0 Bit1=0 Bit0=0
PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);

// Port D initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);

while (1)
      {
      // Place your code here
          for(i=0;i<=9999;i++) //Count From 0 to 9999
          {
            DigitSeperation(i);
            SevenSegment(); //Display Number On 7-Segment
          }
      }
```

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```
71          }
72      }
73  void DigitSeperation(int i)
74  {
75      digit[0] = i % 10;
76      digit[1] = (i / 10) % 10;
77      digit[2] = (i / 100) % 10;
78      digit[3] = (i / 1000) % 10;
79  }
80  void SevenSegment()
81  {
82      PORTB = 0b00001110;
83      PORTC = sevenseg[digit[0]];
84      delay_ms(1);
85
86      if(i>9) //If Number Is 2digt Turn On Its 7-Segment
87      {
88          PORTB = 0b00001101;
89          PORTC = sevenseg[digit[1]];
90          delay_ms(1);
91      }
92
93      if(i>99) //If Number Is 3digt Turn On Its 7-Segment
94      {
95          PORTB = 0b00001011;
96          PORTC = sevenseg[digit[2]];
97          delay_ms(1);
98      }
99
100     if(i>9999) //If Number Is 4digt Turn On Its 7-Segment
101     {
102         PORTB = 0b00000111;
103         PORTC = sevenseg[digit[3]];
104         delay_ms(1);
105     }
106 }
```

